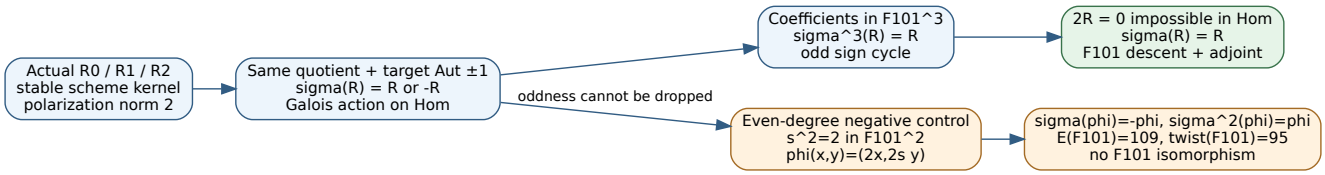
Prime-field descent through an odd coefficient field

2026-11-12 · actual R0/R1/R2 · scientific standing unsigned

The accepted Richelot maps admit a second global F101 descent proof. Their correspondence coefficients lie in $F101^3$ even though evaluating geometric points uses $F101^6$. Stable scheme kernels and polarized target automorphisms±1 leave a sign ambiguity. The odd coefficient degree rules out the negative sign in the torsion-free Hom group. Existing rational odd-witness proofs and all original source controls remain unchanged. Written proof ODD_FIELD_DESCENT_PROOF.md/PCC6b3ff6. [pcc:unverified]



Exact theorem and geometric bridge

Let A,B be principally polarized abelian varieties over Fq and let a nonzero isogeny R:A→B be defined over Fq^m for odd m. Assume its finite scheme kernel is Galois stable, its polarization norm is an integer n, and every geometric origin-preserving polarized automorphism of B is±1. Then R and its polarized adjoint descend to Fq. The field degree and automorphism assumptions are essential parts of this statement. [pcc:unverified]

Conjugation gives another quotient with the same scheme kernel, so sigma(R)=aR. Equal polarization pullbacks make a polarized: pullback by an isogeny is injective on polarization homomorphisms. Thus a=±1. If sigma(R)=-R, odd iteration gives sigma^m(R)=-R; the coefficient field also gives sigma^m(R)=R. Hence2R=0, contradicting nonzero R in torsion-free Hom. Finite Galois descent then supplies the base-field morphism. This sigma is the Galois action on homomorphisms, not an inseparable Frobenius morphism, and point groups are not torsion-free. [pcc:unverified]

Actual instantiation

The cubic constructor forms the quadratic factors and bracket/determinant duals in $F101[T]/(T^3+6T^2+13T+56)$, then embeds with zero second tower component. Sources, targets and ordinate scales22/47/50 lie in F101. The retained complete branch certificates check all120 triple-determined PGL2 candidates for each T0/T1/T2; hyperelliptic theory and strong Torelli give the required polarized Aut±1. Apply m3,n2. The separate HLP-map descent arguments are unchanged; no Aut±1 assumption is added to C1/C2. [pcc:unverified]

Why the odd-degree hypothesis matters

A concrete even-degree counterexample

Take E:$y^2=x^3+x+46$ and E′:$y^2=x^3+4x+65$ over F101. With $s^2=2$, phi(x,y)=(2x,2s y) is an origin-preserving isomorphism over $F101^2$. Its kernel is trivial and stable, and the generic j12 target has polarized Aut±1. But sigma(s)=-s gives sigma(phi)=-phi and $sigma^2$(phi)=phi. This is a global equality of displayed maps. Unequal rational point counts109/95 exclude any F101 isomorphism. [pcc:unverified]

Existing-engine native control01/PCCcd2506 counts both groups, checks all109 source images and $all109^2$ additions, and finds negative Frobenius and nonrationality for every108 nonzero image. It uses the existing flat field Fp[s]/(s^2-2) and existing elliptic operations; no new arithmetic engine or timing benchmark. This preserves an actual counterexample to dropping oddness rather than inferring an obstruction from failed search. [pcc:unverified]

Formal proof map and limits

OddFieldDescent.odd_sign_descent and cubic_sign_descent prove the faithful module implication with explicit negation, fixed odd iterate, nonzero and no2-torsion hypotheses, PCCfdec11/94f071. Three iterate lemmas have individual records39f845/00162f/9c5016. The generic ring identity quadratic_twist_curve/PCC8088e8 checks the coordinate scaling $s^2=d$. All six written statements have bound theorem records and local pass verdicts; scientific standing remains unsigned. [pcc:unverified]

The final Lean4.35.0 gate audits six written theorems plus three generated iterate equations, zero trusted native. The project source census is131 written declarations; generated equations are excluded. Complete standalone correctness02 passes9tests and strictClippy. PCCb33222 pins these receipts. CI now includes22 named Lean modules and the ninth standalone test; their next exact-head CI remains a separate obligation. [pcc:unverified]

Actual fields, abelian varieties, Galois action, regular quotient, polarization, Hom torsion-freeness, Torelli and finite Galois descent remain written bridges. The formalization is partial; it does not claim constructed Lean geometry. This result changes no curve, normalization, catalog, DLP algorithm or measured cost. Existing transfer graph sources remain current because no map or edge changed. [pcc:unverified]

Sources and reproducibility

Primary descent source: Milne, Abelian Varieties v2(2008), ChapterIII Proposition1.13(a), finite-Galois equivalence for quasi-projective varieties; full faithfulness descends invariant morphisms. Checked PDF SHA256 `f5ca4e63e5092a4b102daad1470e4cbed5fe8f82115e3a28c8881e3f67f6aaef`. Strong Torelli: Milne, Jacobian Varieties(2021), Theorem12.1, checked in the retained chain proof. Actual coefficients and branch certificates are the existing consumer sources, with full citations in the written proof. [pcc:unverified]

Reproduce with `lean/OddFieldDescent.lean`, the ordinary proof gate and `successor-preflight/src/odd_descent.rs`. Receipts and exact sources are in `evidence/odd-descent-*` and `evidence/even-field-negative-*`. KB/atlas access remains unavailable; no novelty or absence claim is made.